

Discrete Probability Distributions

Binomial Distribution

Binomial distribution is a discrete probability distribution and has only two outcomes success or failure.

All its trials are independent.

The pmf is given by:

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

p is the probability of success and $1 - p$ is the probability of failure.

Functions for binomial distribution. There are four functions for handling binomial distribution:

1. Probability Density Function `dbinom(x, size, prob)`
2. Cumulative Distribution Function `pbinom(x, size, prob, lower.tail=FALSE)`
3. Quantile Function `qbinom(p, size, prob)`
4. Generating random numbers `rbinom(n, size, prob)`

where:

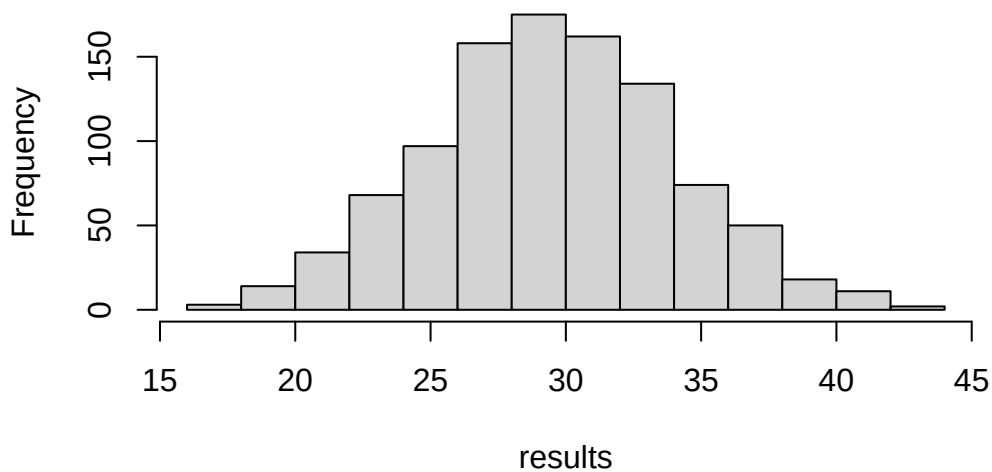
- x is a vector of numbers
- p is a vector of probabilities
- n is number of observations
- $size$ is the number of trials
- $prob$ is the probability of success of each trial

Examples - Generating Random Numbers (rbinom) Example 1

Generate a vector that shows the number of successes of 1000 binomial experiments with 100 trials where the probability of success in each trial is 0.3.

```
results <- rbinom(1000, size=100, prob=0.3)
hist(results)
```

Histogram of results



Example 2

Generate a vector that shows the number of successes of 10 binomial experiments with 100 trials where the probability of success on each trial is 0.5.

Examples - `dbinom()` This function gives the probability density distribution at each point.

Example 1

Bob makes success in 60% of his free-throw attempts. If he shoots 12 free throws, what is the probability that he achieves exactly 10 success?

```
dbinom(x=10, size=12, prob=.6)
```

```
[1] 0.06385228
```

Example 2

Subha flips a fair coin 20 times. What is the probability that the coin lands on heads exactly 7 times?

```
dbinom(x=7, size=20, prob=.5)
```

```
[1] 0.07392883
```

Examples - `pbinom()` This function gives the cumulative probability of an event. It is a single value representing the probability.

Example 1

Raju flips a fair coin 5 times. What is the probability that the coin lands on heads more than 2 times?

```
pbinom(2, size=5, prob=.5, lower.tail=FALSE)
```

```
[1] 0.5
```

Example 2

Suppose a bowler scores a strike on 30% of his attempts when he bowls. If he bowls 10 times, what is the probability that he scores 4 or fewer strikes?

```
pbinom(4, size=10, prob=0.3)
```

```
[1] 0.8497317
```

Examples - qbinom() This function takes the probability value and gives a number whose cumulative value matches the probability value.

Example 1

Find the 10th quantile of a binomial distribution with 10 trials and probability of success on each trial = 0.4

```
qbinom(0.10, size=10, prob=0.4)
```

```
[1] 2
```

Example 2

Find the 40th quantile of a binomial distribution with 30 trials and probability of success on each trial = 0.25

```
qbinom(0.40, size=30, prob=0.25)
```

```
[1] 7
```

Geometric Distribution

Geometric distribution is a type of discrete probability distribution that represents the probability of the number of successive failures before a success is obtained in a Bernoulli trial.

The distribution is based on three assumptions:

- The trials being conducted are independent.
- There can only be two outcomes of each trial - success or failure.
- The success probability, denoted by p , is the same for each trial.

Examples - dgeom() The dgeom function finds the probability of experiencing a certain amount of failures before experiencing the first success in a series of Bernoulli trials

Syntax is **dgeom(x,prob)**

where:

x number of failures before a success

prob probability of success on a given trial

Example 1

A researcher is waiting outside of a library to ask people if they support a certain law. The probability that a given person supports the law is $p = 0.2$. What is the probability that the fourth person the researcher talks to is the first person to support the law?

```
dgeom(x=3, prob=.2)
```

```
[1] 0.1024
```

Examples - pgeom() The pgeom function finds the probability of experiencing a certain amount of failures or less before experiencing the first success in a series of Bernoulli trials, using the following syntax:

pgeom(q,prob)

where:

q number of failures before first success

prob probability of success on a given trial

Example 1

A researcher is waiting outside of a library to ask people if they support a certain law. The probability that a given person supports the law is $p = 0.2$. What is the probability that the researcher will have to talk to more than 5 people to find someone who supports the law?

```
1 - pgeom(q=5, prob=.2)
```

```
[1] 0.262144
```

Examples - qgeom() The qgeom function finds the number of failures that corresponds to a certain percentile, using the following syntax: **qgeom(p, prob)**

where:

p: percentile

prob: probability of success on a given trial

Example 1

A researcher is waiting outside of a library to ask people if they support a certain law. The probability that a given person supports the law is $p = 0.2$. We will consider a "failure" to mean that a person does not support the law. How many "failures" would the researcher need to experience to be at the 90th percentile for number of failures before the first success?

```
qgeom(p=.90, prob=0.2)
```

```
[1] 10
```

Examples - rgeom() The rgeom function generates a list of random values that represent the number of failures before the first success, using the following syntax:

```
rgeom(n, prob)
```

where:

n: number of values to generate

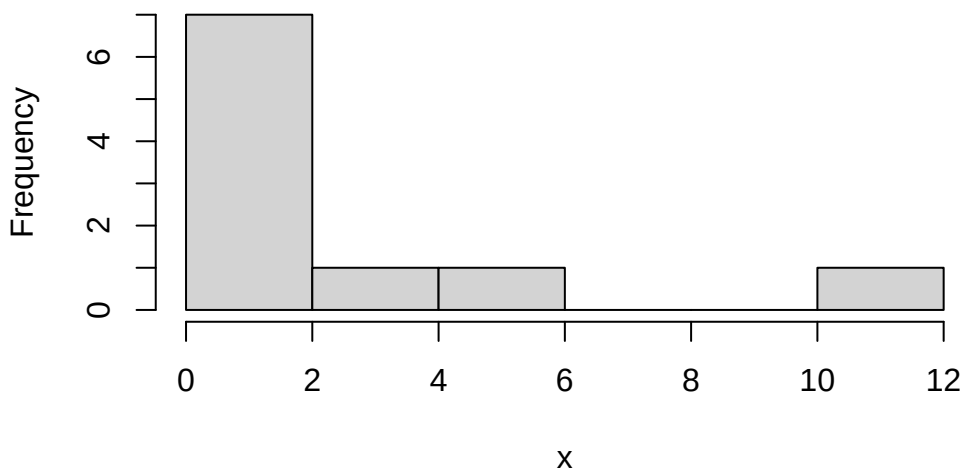
prob: probability of success on a given trial

Example 1

A researcher is waiting outside of a library to ask people if they support a certain law. The probability that a given person supports the law is $p = 0.2$. We will consider a “failure” to mean that a person does not support the law. Simulate 10 scenarios for how many “failures” the researcher will experience until she finds someone who supports the law.

```
x <- rgeom(n=10, prob=.2)
hist(x)
```

Histogram of x



Poisson Distribution

Poisson Distribution deals with the probability distribution of data values taking the mean into consideration.

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Examples - qpois() The R qpois function allows obtaining the corresponding Poisson quantiles for a set of probabilities. The qpois function finds the number of successes that corresponds to a certain percentile based on an average rate of success.

The syntax is

```
qpois(p, lambda, lower.tail = TRUE)
```

Example 1

It is known that a certain website makes 10 sales per hour. How many sales would the site need to make to be at the 90th percentile for sales in an hour?

```
qpois(p=.90, lambda=10)
```

```
[1] 14
```

Examples - rpois() To draw n observations from a Poisson distribution the rpois function can be used. The following block of code summarizes the arguments of the function.

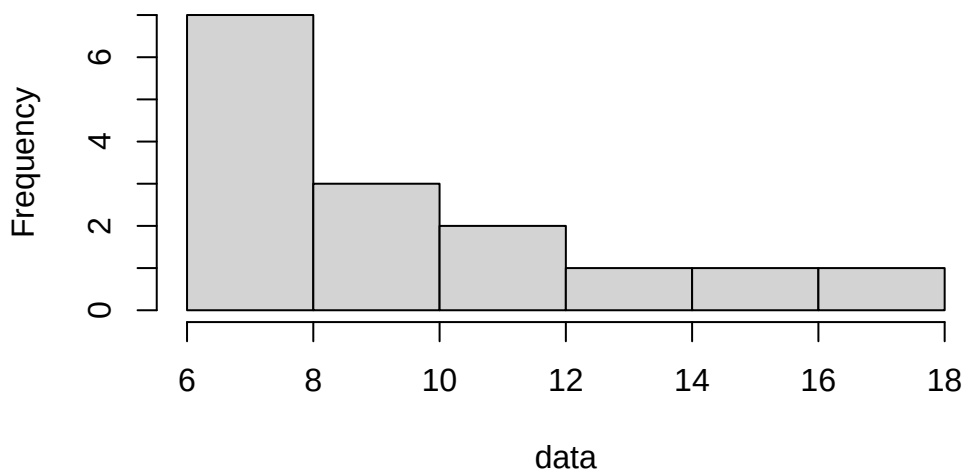
```
rpois(n, lambda)
```

Example 1

Generate a list of 15 random variables that follow a Poisson distribution with a rate of success equal to 10.

```
data <- rpois(n=15, lambda=10)
hist(data)
```

Histogram of data



Example 2 - ppois

If there are twelve cars crossing a bridge per minute on average, find the probability of having seventeen or more cars crossing the bridge in a particular minute.

```
ppois(16, lambda=12, lower=FALSE)
```

```
[1] 0.101291
```

Example 3 - ppois

Consider that the number of visits on a web page is known to follow a Poisson distribution with mean 15 visits per hour. What is the probability of getting

1. 10 or less visits per hour?
2. The probability of getting more than 20 visits per hour
3. The probability of getting 10 to 20 visits per hour.

Solution

```
ppois(10, lambda = 15)
```

```
[1] 0.1184644
```

```
ppois(20, lambda = 15, lower.tail = FALSE)
```

```
[1] 0.08297091
```

```
ppois(20, lambda = 15) - ppois(10, lambda = 15)
```

```
[1] 0.7985647
```

Continuous Probability Distributions

Exponential Distribution

The pdf is given as:

$$f(x) = \lambda e^{-\lambda x}$$

To simulate observations using exponential distribution use the function `rexp(n, rate)`. The rate is given as $rate = \frac{1}{\lambda}$

```
data <- rexp(1000, rate = 0.5)
hist(data)
```